

Project Cost Management (Earned Value Management)

SECTION: Key Processes

1. **Plan Cost Management** — how costs will be estimated, budgeted, and controlled.
2. **Estimate Costs** — approximating the cost of resources (analogous estimating, parametric estimating, bottom-up estimating, three-point estimating).
3. **Determine Budget** — aggregating estimated costs into an authorized **cost baseline** (a time-phased budget, often shown as an S-curve).
4. **Control Costs** — monitoring cost performance, managing changes to the cost baseline. This is where **Earned Value Management (EVM)** is applied.

SECTION: Core EVM Terms

- **PV (Planned Value)**: the authorized budget assigned to scheduled work as of a point in time — "what should have been spent by now."
- **EV (Earned Value)**: the value of work actually completed, expressed in terms of the approved budget — "what the completed work is worth."
- **AC (Actual Cost)**: the actual cost incurred for the work completed — "what was really spent."
- **BAC (Budget at Completion)**: the total planned budget for the entire project.

SECTION: EVM Formulas

| Metric | Formula | Interpretation |

|---|---|---|

| Cost Variance (CV) | $EV - AC$ | Positive = under budget; Negative = over budget |

| Schedule Variance (SV) | $EV - PV$ | Positive = ahead of schedule; Negative = behind schedule |

| Cost Performance Index (CPI) | EV / AC | >1 = efficient cost performance; <1 = cost overrun |

| Schedule Performance Index (SPI) | EV / PV | >1 = ahead of schedule; <1 = behind schedule |

| Estimate at Completion (EAC) | BAC / CPI | Forecasted total cost if current cost performance continues |

| Estimate to Complete (ETC) | $EAC - AC$ | Expected cost of remaining work |

| Variance at Completion (VAC) | $BAC - EAC$ | Projected budget surplus/deficit at project end |

| To-Complete Performance Index (TCPI) | $(BAC - EV) / (BAC - AC)$ | Efficiency required for remaining work to meet BAC |

SECTION: Worked Example

A project has $BAC = \$100,000$. At the review date: $PV = \$40,000$, $EV = \$35,000$, $AC = \$45,000$.

- $CV = 35,000 - 45,000 = -\$10,000$ (over budget)
- $SV = 35,000 - 40,000 = -\$5,000$ (behind schedule)
- $CPI = 35,000 / 45,000 = 0.78$ (getting only \$0.78 of value per \$1 spent)
- $SPI = 35,000 / 40,000 = 0.875$ (behind schedule)
- $EAC = 100,000 / 0.78 \approx \$128,205$ (forecast final cost, higher than BAC — bad news)

SECTION: Estimating Techniques

- **Analogous Estimating:** uses historical data from similar past projects; fast but less accurate.
- **Parametric Estimating:** uses a statistical relationship between historical data and variables (e.g. cost per square meter, cost per line of code).
- **Bottom-Up Estimating:** estimates individual work packages/activities and aggregates them; most accurate but time-consuming.
- **Three-Point Estimating (PERT-based cost):** $(Optimistic + 4 \times \text{Most Likely} + Pessimistic) / 6$ — accounts for uncertainty.

SECTION: Cost Baseline vs Budget vs Funding Requirements

- **Cost Baseline:** the approved, time-phased budget used to measure performance (excludes management reserve).
- **Project Budget:** cost baseline + management reserve.
- **Funding Requirements:** often released periodically/incrementally rather than as one lump sum, sometimes exceeding the baseline briefly to allow for expenditure spikes (funding "steps" above the S-curve).

SECTION: Common Exam Angles

- Calculate CV, SV, CPI, SPI, EAC given PV/EV/AC/BAC — this is a very common numeric question type.
- Interpret whether a project is ahead/behind schedule and under/over budget from index values.
- Explain the difference between cost baseline and project budget.
- Choose the appropriate estimating technique for a scenario (early rough estimate vs detailed estimate needed).